Homework250926:

Question: What is Python request and what does it do?

My understanding is that python request is meant to simplify working between multiple different applications and to manage sessions where a developer is integrating, patching, deleting, posting or getting. The request function seems to also handle errors and to provides support for JSON usage.

Question: XML and JSON

My understanding is that JSON is for lightweight data exchange and usage in instances like chatbots where data is presented as key-value pairs or as arrays. Application configuration files are also often used as JSON.

Advantages	Disadvantages
Compact	Not suitable for more complex files
Works with almost all modern coding languages	Limited data types
Human readable	You cannot comment without adding metadata
Flexible design	Third party tools have to enforce schemas/structures

XML is a markup language that defines a set of rules for encoding documents in a format that is both human- and machine-readable.

Advantages	Disadvantages
Enforces strict data typing, structural rules, and integrity checks before processing data	Steeper Learning Curve
Human readable	Repeating tag names for every single data record or array item adds unnecessary data bloat.
Extensive and extendable with space for hierarchies	Processes/parses slower than JSON
It integrates with transformation to reshape data structures into HTML, PDF, or other formats.	Much higher bandwidth consumption than JSON

Question: RESTful API

RESTful API is an architectural system that allows developers to network apps and to build communication channels between clients using HTTP. It's used for third party integrations and for connecting frontends and backends. It operates on a client-server model.

Advantages	Disadvantages
The client and server are completely decoupled and this reduces risk and increases interoperability	Over-fetching or under-fetching of data is common
Language Agnostic	Not efficient for real-time streaming (compared to other tools)
Cacheable	Every request must be independent, so authentication tokens or user context must be re-transmitted with every single call
Very scalable	Complex applications need numerous distinct endpoints for different resources and relationships, which can become difficult to manage and document.